

Ryan Cho

Senior Software Engineer | Cloud & AI Platforms

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PROFESSIONAL SUMMARY

Software engineer with 8+ years delivering cloud-native APIs, distributed services, and production AI capabilities for finance and enterprise workflows. Designed retrieval pipelines, agent orchestration, and multi-provider LLM abstractions while owning observability, CI/CD, and reliability across high-throughput platforms. Proficient in Python and Java with a track record of optimizing latency, cost, and accuracy in mission-critical production systems.

WORK EXPERIENCE

Software Engineer

May/2022 - May/2026

Aspen Capital

Remote | 10th Floor, New York

- Architected **cloud-native microservices** in Python and Java supporting portfolio analytics, trade operations, and client reporting across a distributed platform serving internal teams and external partners.
- Built **production RAG pipelines** over fund documentation and market research corpora, combining vector search, metadata filtering, and citation-aware responses to accelerate analyst research workflows.
- Designed **agent orchestration workflows** with tool invocation patterns for document extraction, compliance checks, and structured data enrichment, reducing manual review cycles for operations staff.
- Implemented a **multi-provider LLM abstraction layer** spanning OpenAI, Anthropic, and Vertex AI, enabling model routing, fallback policies, and centralized prompt governance without vendor lock-in.
- Established **evaluation frameworks** measuring retrieval precision, answer faithfulness, latency percentiles, and cost per request, giving engineering and business stakeholders shared quality benchmarks.
- Deployed containerized services on **Kubernetes with CI/CD automation**, infrastructure-as-code provisioning, and blue-green releases that maintained uptime during frequent model and API upgrades.
- Instrumented **end-to-end observability** with distributed tracing, structured logging, and SLO dashboards across inference, retrieval, and downstream API dependencies in production environments.
- Optimized **AI workload performance and spend** through caching, batching, token budgeting, and selective model tiering, improving response times while controlling inference costs at scale.
- Partnered with portfolio managers, risk teams, and client engineering groups to **integrate AI capabilities into existing enterprise APIs**, aligning technical design with regulated finance workflows.

Technology Analyst, Incoming Technology Analyst, Technology

May/2019 - Apr/2022

Summer Analyst,

BNP Paribas

Remote | Seventh Avenue, New York

- Developed **Java and Python backend services** for trading, risk, and reporting platforms, exposing REST APIs consumed by front-office applications and downstream batch processing systems.
- Contributed to **microservices decomposition** of legacy monoliths, defining service boundaries, event-driven interfaces, and contract tests that improved deployment independence across global teams.
- Built **serverless and container-based workloads** on AWS for data ingestion, transformation, and scheduled reconciliation jobs supporting capital markets operations.
- Owned **CI/CD pipeline enhancements** with automated testing gates, artifact promotion, and environment promotion workflows that shortened release cycles for critical production services.
- Implemented **production observability** using metrics, alerting, and runbooks for latency, error rates, and queue backlogs across high-volume market data and settlement pipelines.
- Diagnosed and resolved **production performance bottlenecks** in database queries, message consumers, and API gateways, restoring throughput during peak trading sessions.
- Collaborated with quants, business analysts, and infrastructure engineers on **enterprise system integration** spanning internal platforms, vendor feeds, and regulatory reporting interfaces.

- Mentored summer analysts and incoming hires on **service design, code review practices**, and safe production change management within a regulated banking environment.

Software Engineering Intern

Jun/2018 - May/2019

Chexology

Remote | 17th Floor, New York

- Shipped **full-stack features** in React and Node.js for a restaurant technology platform, delivering order management screens and admin tooling used by pilot merchant locations.
- Implemented **REST API endpoints** with validation, authentication middleware, and pagination patterns that supported mobile clients and internal operations dashboards.
- Integrated **third-party payment and notification services**, handling webhook retries, idempotency keys, and error surfacing to improve checkout reliability.
- Containerized application components with **Docker** and supported staging deployments, gaining early exposure to environment parity and release checklists.
- Participated in **code reviews and sprint planning**, translating product requirements into technical tasks and iterating quickly based on merchant feedback.
- Added **unit and integration tests** around critical checkout flows, reducing regressions as the team expanded feature scope across multiple service modules.

Web Developer

Sep/2017 - Nov/2017

Bayesquare Foundation

Remote | New York

- Built **responsive web pages** for a nonprofit foundation site using HTML, CSS, and JavaScript, improving content accessibility across desktop and mobile visitors.
- Developed **server-side form handling** and lightweight CMS updates that allowed non-technical staff to publish events and donation campaigns without engineering support.
- Applied **SEO and performance basics** including semantic markup, image optimization, and page load tuning for community outreach landing pages.
- Collaborated with stakeholders to **translate design mockups into production-ready UI**, iterating on layout and typography based on user feedback.
- Maintained **version-controlled codebase hygiene** with structured commits and deployment checklists during a fast-paced launch window.

EDUCATION

New York University

Bachelor of Arts (B.A.), Computer Science

2016-2020

SKILLS

Frontend: React, TypeScript, JavaScript, HTML/CSS, Next.js, Responsive UI

Backend: Python, Java, FastAPI, Spring Boot, Node.js, REST APIs, Microservices, Serverless

Databases: PostgreSQL, Redis, Elasticsearch, MongoDB, Vector Search (pgvector, Pinecone)

Cloud & DevOps: AWS, Docker, Kubernetes, Terraform, GitHub Actions, Jenkins, IaC

Monitoring & Tools: Datadog, Prometheus, Grafana, OpenTelemetry, OpenAI API, Anthropic API, Vertex AI, LangChain

Workflow & PM: Agile/Scrum, Jira, Technical Design, Stakeholder Collaboration, Code Review, Mentorship